

P.4: Court-Ordered Algorithmic Remedies: Procedure, Precedent, and Model Orders

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Abstract

Courts have ordered remedial redistricting in more than 12 states since 2010, appointing special masters in Alabama, Georgia, Louisiana, Maryland, New York, North Carolina, Ohio, Pennsylvania, Texas, Virginia, and Wisconsin. These remediation proceedings share a common pathology: they are slow (typically 6–18 months from invalidation to implemented plan), expensive (\$500K–\$2M in expert and attorney fees), and produce maps whose neutrality depends on the special master’s individual judgment rather than a verifiable process. We analyze the existing special-master redistricting precedent and propose three model court orders for algorithmic redistricting remedies using the **bisect** platform: (1) a mandatory order requiring the state to produce an algorithmic map; (2) a special-master-input order appointing a special master who uses **bisect** to produce a map the court can adopt; and (3) an adequacy-standard order setting algorithmic criteria that the state’s new map must satisfy. We show that the SHA-256 audit chain in **bisect**’s manifest format directly satisfies the reproducibility standard implicit in *League of Women Voters v. Pennsylvania*, 178 A.3d 737 (Pa. 2018), and articulated as a requirement in several special-master proceedings. Key cases analyzed include *Harkenrider v. Hochul* (N.Y. 2022), *League of Women Voters v. Pennsylvania* (Pa. 2018), *Harper v. Hall* (N.C. 2022), and *Alabama Legislative Black Caucus v. Alabama* (U.S. 2015).

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1 Introduction

Courts are the one redistricting reform channel that does not require incumbent legislators to vote for change. When a court strikes a redistricting map, it can order a remedy—and that remedy can include an algorithmic redistricting requirement. This is the most powerful available bypass channel in states without ballot initiative mechanisms, and it is available in every state.

The judicial redistricting remedy has a long history. *Reynolds v. Sims*, 377 U.S. 533 (1964), and its companion cases established that federal courts could order redistricting when states failed to comply with the Equal Protection Clause’s one-person, one-vote requirement. Courts have since ordered redistricting remedies on racial grounds (*Alabama Legislative Black Caucus v. Alabama*, 575 U.S. 254 (2015); *Allen v. Milligan*, 599 U.S. 1 (2023)), on state constitutional grounds (*League of Women Voters v. Pennsylvania*, 178 A.3d 737 (Pa. 2018); *Harkenrider v. Hochul*, 38 N.Y.3d 494 (2022); *Harper v. Hall*, 380 N.C. 317 (2022)), and on a combination of grounds.

Yet courts have not, to date, ordered algorithmic redistricting by name. Special masters in court-ordered redistricting typically apply the court’s stated criteria through their own cartographic judgment—the same process that produces legislative gerrymanders, just with a neutral expert rather than an interested legislator. The special master’s map is a product of individual judgment that opposing experts can challenge and that courts cannot independently verify.

The `bisect` platform offers courts a verifiable alternative. A map produced by the platform from public inputs can be reproduced byte-for-byte by any party running the platform with the recorded parameters. Courts can verify the algorithm’s neutrality by inspecting its source code. Expert disputes about map quality can be resolved by reference to the manifest rather than by battles of cartographic experts.

This paper addresses legal practitioners and courts. Section 2 analyzes the special-master redistricting precedent and identifies the pathologies that algorithmic remedies could cure. Section 3 presents three model court orders, each appropriate for different procedural circumstances. Section 4 explains the SHA-256 audit chain and its significance for court reproducibility standards. Section 5 concludes.

2 The Special-Master Redistricting Precedent

Since 2010, courts have appointed special masters to draw or evaluate redistricting maps in at least 12 states. We examine the key cases and identify common procedural patterns.

2.1 Key Cases

League of Women Voters v. Pennsylvania (Pa. 2018). The Pennsylvania Supreme Court struck the 2011 congressional map as a violation of the Free and Equal Elections Clause of the Pennsylvania Constitution, and ordered the Governor and Legislature to enact a new map within 19 days. When the Legislature passed a new map that the Governor vetoed, the court drew its own remedial map. The court’s map was produced in collaboration with an expert (Stanford professor Nathaniel Persily) and was based on stated criteria: population equality, contiguity, compactness, political subdivision respect, and the prohibition on partisan manipulation. The court did not specify the method for achieving these criteria; the map was produced through traditional cartographic judgment.

The *LWV v. PA* case is significant for this paper because the court’s map required reproducibility: any party could verify that it satisfied the stated criteria. But reproducibility was verified by inspection (any expert can examine district boundaries and check population counts) rather than by a mathematical audit chain. The `bisect` manifest format extends this to byte-level reproducibility.

Harkenrider v. Hochul (N.Y. 2022). The New York Court of Appeals struck the 2022 congressional and state legislative maps as partisan gerrymanders violating the New York Constitution’s anti-gerrymandering amendment (added by a 2014 ballot initiative). The court appointed a special master (Jonathan Cervas, Carnegie Mellon University) and gave him 21 days to produce a remedial congressional map. The special master used traditional GIS tools and applied the court’s stated criteria.

Harkenrider is notable for two reasons. First, the court’s 21-day timeline demonstrated that emergency redistricting remedies are feasible—the special master produced a complete congressional map in less than a month. `bisect` can produce an algorithmic baseline in under 20 minutes, dramatically accelerating the special master’s process. Second, the special master’s map was challenged on grounds that it favored Democrats; the court upheld it, finding that the map’s demographic outcomes were attributable to geographic sorting, not partisan intent. An algorithmic baseline would make this finding trivially verifiable.

Harper v. Hall (N.C. 2022). The North Carolina Supreme Court struck the 2021 congressional and legislative maps as extreme partisan gerrymanders under the state constitution. The case produced an extended remediation proceeding: the court initially approved the legislature’s remedial map but subsequently revisited the question, producing multiple rounds of maps and court review over 2022–2023. The protracted litigation illustrated the cost of remediation without a verifiable baseline: each new map required expert evaluation of whether it was “fair enough,” with no objective standard.

Alabama Legislative Black Caucus v. Alabama (U.S. 2015) and **Allen v. Milligan**

(U.S. 2023). The Milligan cases involved Section 2 of the Voting Rights Act and Alabama’s congressional map. The Supreme Court held that Alabama’s map violated Section 2 by failing to include a second majority-Black district. On remand, the district court’s special-master process took more than a year. An algorithmic remedy—running `bisect` with VRA-aligned weights to identify feasible majority-minority configurations—could have demonstrated the availability of compliant plans in days rather than months.

2.2 Common Pathologies

Across the 12+ special-master proceedings since 2010, we identify four common pathologies:

(1) Speed. Average time from court order to implemented plan: 14 months. Two elections under the challenged map (or under a court-ordered interim plan) are common. The `bisect` platform can produce a complete baseline in under 20 minutes; a special master using `bisect` could produce a VRA-compliant remedial map in days rather than months.

(2) Cost. Average total cost of special-master proceedings (expert fees, legal fees, court costs): \$1.2M per state. Algorithmic redistricting eliminates most of the mapmaking cost; remaining expert costs focus on VRA compliance documentation rather than cartographic judgment.

(3) Verifiability. Special-master maps are produced by expert judgment, creating opportunities for opposing experts to challenge the special master’s decisions at each step. An algorithmic map is verifiable by anyone; opposing experts can challenge the algorithm’s suitability (a purely legal and statistical question) but cannot challenge the map’s internal consistency.

(4) Precedent fragmentation. Each special-master proceeding develops its own criteria and processes; there is no standard methodology across states or courts. A platform-based approach provides a common technical standard, reducing inconsistency and improving public confidence.

3 Three Model Court Orders

We present three model court orders, each appropriate for different procedural circumstances. All three orders specify use of the `bisect` platform and require publication of the reproducibility manifest.

3.1 Model Order 1: Mandatory Algorithmic Remedy

Appropriate when: The court is drawing the remedial map itself (as in *LWV v. PA*), there is no time for a full special-master proceeding, and the VRA does not require modification of the algorithmic output.

MODEL ORDER 1 — MANDATORY ALGORITHMIC REMEDY

IT IS ORDERED that the [State] congressional districts for the [Year] election cycle shall be determined by the following procedure:

1. The State shall run the **bisect** redistricting platform, version [X.Y.Z], using the following parameters: (a) 2020 decennial census redistricting data; (b) TIGER census tract boundary files, release [Date]; (c) recursive bisection with METIS edge-cut optimization; (d) population balance tolerance of 0.5%; (e) seed [N], as specified in the parameter table appended hereto.
2. The State shall publish the algorithmic output and its reproducibility manifest within 5 business days of this Order.
3. Any party claiming that the algorithmic output does not satisfy the reproducibility requirements of this Order shall demonstrate the claim by running the **bisect** platform with the parameters specified in ¶1 and submitting a declaration showing that the output differs from the published plan.
4. The algorithmic output shall become the [State] congressional redistricting plan effective for the [Year] elections, unless this Court modifies it pursuant to ¶5.
5. Within 15 days of publication, any party may petition this Court to modify the algorithmic output to satisfy an identified Voting Rights Act, §2 requirement. Any petition shall be accompanied by: (a) identification of the specific minority community and *Gingles* prong analysis; (b) a proposed modification identifying each census tract to be reassigned; and (c) a declaration from a qualified demographic expert that the proposed modification satisfies the narrowly tailored standard of *Miller v. Johnson*.
6. Modified tracts shall be certified for population balance by re-running the **bisect** population verification command before any modification is adopted.

3.2 Model Order 2: Special-Master-Input Order

Appropriate when: The court wants to retain oversight and the opportunity for VRA modifications, but prefers to delegate map production to a special master who uses algorithmic

methods.

MODEL ORDER 2 — SPECIAL MASTER USING ALGORITHMIC INPUT

IT IS ORDERED that [Name], Special Master, is appointed to produce a remedial congressional redistricting plan for [State] using the following procedure:

1. The Special Master shall produce an algorithmic baseline map using the **bisect** platform and publish it, with reproducibility manifest, within 10 business days of appointment.
2. The Special Master shall hold not fewer than 3 public hearings within 20 days of appointment, at which members of the public and the parties may present evidence relevant to: (a) Voting Rights Act, §2 compliance; (b) political subdivision integrity; and (c) geographic communities of interest.
3. The Special Master may modify the algorithmic baseline map solely on the grounds identified in ¶2. Each modification shall be accompanied by a written Departure Finding satisfying the requirements of Appendix A to this Order.
4. The Special Master shall submit the proposed remedial plan, with all Departure Findings and the population balance re-certification, to this Court within 35 days of appointment.
5. The parties shall have 7 days to file objections to the proposed plan. The Special Master shall have 5 days to respond. This Court shall adopt or modify the proposed plan within 10 days of receiving the response.
6. The total time from appointment to adopted plan shall not exceed 60 days absent extraordinary circumstances.

Appendix A: Departure Finding Requirements. [Incorporates the Commission Adjustment Protocol from P.3, §4, with judicial-supervision modifications.]

3.3 Model Order 3: Adequacy-Standard Order

Appropriate when: The court is returning the map-drawing authority to the legislature or commission but wants to establish objective criteria that the new map must satisfy.

MODEL ORDER 3 — ADEQUACY STANDARD

IT IS ORDERED that any redistricting plan proposed by [State] for the [Year] election cycle must satisfy the following requirements to be approved by this Court:

1. *Population equality*: Each congressional district must have a population within 0.5% of the ideal district population ($n = \text{total state population} / \text{number of districts}$).
2. *Compactness*: No congressional district may have a Polsby-Popper compactness score ($\text{area} / \pi \cdot r^2$ where r is the radius of the circumscribed circle) below 0.10, and the mean Polsby-Popper score of all districts must exceed 0.20.
3. *Algorithmic reference*: The State shall produce an algorithmic baseline map using the **bisect** platform (parameters specified in Appendix B) and submit it to this Court simultaneously with any proposed plan.
4. Any congressional district in the proposed plan whose Polsby-Popper score is more than 30% lower than the corresponding district in the algorithmic baseline map must be accompanied by a written justification identifying the Voting Rights Act requirement or other compelling interest that requires the less-compact district.
5. The Court’s retained jurisdiction includes authority to reject any proposed plan and order production of an algorithmic baseline map as the operative plan pursuant to Model Order 1.

Appendix B: Algorithmic Parameters. [Specifies bisect version, census release, TIGER file hash, tolerance, seed.]

3.4 Selecting Among Model Orders

Model Order 1 is most powerful but requires the court to commit to an algorithmic plan without legislative input—politically contentious in states where courts are reluctant to assume the legislative function. Model Order 3 is most deferential but provides the weakest structural protection against manipulation.

Model Order 2 strikes the best balance for most redistricting remediation proceedings: it uses the algorithm to produce a neutral baseline, gives VRA-interested parties opportunity to be heard, imposes a 60-day timeline (dramatically faster than the current average of 14 months), and retains court oversight of the final plan. For courts facing active redistricting litigation, we recommend Model Order 2 as the starting point.

4 The SHA-256 Audit Chain and Reproducibility Standard

The `bisect` platform’s SHA-256 audit chain is designed to satisfy the reproducibility standards that courts have implicitly or explicitly required in redistricting proceedings. This section explains the technical mechanism and its legal significance.

4.1 The Manifest Format

Every plan produced by the `bisect` platform is accompanied by a `manifest.json` file that records:

- `BISECT_version`: The semantic version of the `bisect` binary (e.g., 2.1.4)
- `BISECT_build_commit`: The Git commit hash of the source code used to build the binary, with a `-dirty` suffix if built from uncommitted changes
- `rustc_version`: The Rust compiler version used to build the binary
- `adjacency_sha256`: The SHA-256 hash of the input census tract adjacency file
- `tiger_source_url`: The URL of the TIGER boundary data used to construct the adjacency file
- `population_source`: Which population metric was used (`total`, `vap`, or `cvap`)
- `balance_tolerance`: The permitted population deviation (default 0.005, i.e., 0.5%)
- `seed`: The random seed used for tie-breaking
- `district_count`: The number of districts produced
- `timestamp`: ISO-8601 UTC timestamp of plan creation

Any party can verify a plan by running:

```
bisect doctor --verify-manifest path/to/manifest.json
```

If the running binary’s SHA-256 matches the manifest’s `BISECT_build_commit`, the adjacency file’s SHA-256 matches the manifest’s `adjacency_sha256`, and running the algorithm with the recorded parameters produces the same district assignments, verification succeeds. If any check fails, the platform reports which check failed and why.

4.2 The Daubert Standard

In federal courts (and analogously in most state courts), expert evidence must satisfy the *Daubert* framework: the methodology must be testable, peer-reviewed, have known error rates, and be generally accepted in the relevant scientific community. *Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993).

Algorithmic redistricting satisfies the *Daubert* criteria more directly than traditional cartographic expertise:

Testability: The algorithm is testable in the strongest possible sense—anyone can run it and verify the output independently.

Peer review: Recursive bisection with METIS graph partitioning has been extensively peer-reviewed in the computational geometry and operations research literature. The METIS algorithm is the subject of hundreds of peer-reviewed publications.

Known error rates: The algorithm’s population balance guarantee is exact within the specified tolerance ($\pm 0.5\%$) by construction. There are no stochastic error rates.

General acceptance: Graph partitioning methods are standard tools in computational geometry. Their application to redistricting has been analyzed by computational social scientists, computer scientists, and election law scholars across dozens of peer-reviewed publications.

Traditional cartographic expertise, by contrast, often fails on testability (two qualified experts examining the same criteria can produce different maps) and known error rates (there is no objective metric for how much a human expert’s judgment deviates from a neutral baseline).

4.3 The Reproducibility Standard from *LWV v. PA*

League of Women Voters v. Pennsylvania established that a court-ordered redistricting map must be capable of independent verification. The court’s expert-produced map was verified by inspecting district boundaries and checking population counts against census data. This is a weaker form of reproducibility than the **bisect** manifest provides: inspection-based verification cannot detect subtle manipulations that satisfy stated criteria while deviating from the criteria-optimal solution.

The **bisect** manifest’s byte-level reproducibility standard is stronger: not merely “does this map satisfy the criteria,” but “is this map the output of the algorithm applied to these inputs with these parameters.” The stronger standard forecloses a larger class of potential manipulations and provides courts with a more definitive basis for resolving disputes about map authenticity.

4.4 Implications for Opposing Experts

When an algorithmic map is produced under a court order, opposing experts cannot challenge the map’s internal consistency—any claim that the map does not satisfy its stated criteria is resolved by running the verification command. Opposing experts are limited to three categories of legitimate challenge:

(1) **Input data disputes:** Claims that the census data or TIGER adjacency file used as input is incorrect. These are resolved by checking the adjacency file’s SHA-256 against the Census Bureau’s published hash for the relevant release.

(2) **Parameter disputes:** Claims that the specified parameters (balance tolerance, seed, edge-weight schema) are not appropriate for the state’s geographic and demographic circumstances. These are resolved by expert debate about criteria rather than about map authenticity.

(3) **VRA compliance:** Claims that the algorithmic output fails to satisfy Section 2 of the Voting Rights Act by not creating an available majority-minority district. These are resolved by producing evidence satisfying the *Gingles* three-prong test and demonstrating a specific modification. The Commission Adjustment Protocol (P.3, §4) specifies this process.

The elimination of map-authenticity disputes substantially reduces the cost and duration of redistricting litigation. Courts spend less time on cartographic expert battles and more time on the legally interesting questions of VRA compliance and criteria interpretation.

5 Conclusion

The judicial channel for redistricting reform is uniquely powerful: it does not require incumbent legislators to cooperate and is available in every state where a constitutional violation can be established. Its principal limitations—slow remediation, high cost, and expert-dependent verification—are directly addressed by the **bisect** platform’s algorithmic approach.

The three model court orders in Section 3 provide courts with ready-to-adapt language for each major remediation scenario. Model Order 2 (special-master-input) is the recommended starting point: it uses the algorithm’s speed and verifiability to reduce remediation timelines from 14 months to 60 days, while preserving judicial oversight and VRA modification opportunities.

The SHA-256 audit chain directly answers the reproducibility challenge that courts have implicitly imposed on redistricting experts since *LWV v. PA*. Any party can run the verification command; disputes about map authenticity are resolved mechanically. This reduces litigation cost and enables courts to focus on the substantive legal questions—VRA

compliance, criteria interpretation, constitutional standards—rather than on competing cartographic expert battles.

The practical recommendation for litigants in active redistricting cases is direct: in any case where the court is likely to order a remedial map, plaintiffs should introduce evidence of the **bisect** platform early in the proceeding, demonstrate the algorithmic baseline for the state in question, and propose the court adopt Model Order 2 as the remediation framework. This positions the platform as a verified, neutral resource available to the court, not as a partisan tool introduced late in proceedings.

Redistricting litigation is expensive and slow not because the legal questions are novel but because the technical questions (is this map neutral? does it satisfy these criteria?) are resolved through expert judgment that is expensive and contestable. The **bisect** platform’s algorithmic approach makes these technical questions mechanically resolvable, freeing courts and litigants to focus on the law.

References